

BOL'SHAYA MURTA, Russian Federation

WMO: 294710

Lat: 56.8950N

Lon: 93.1650E

Elev: 590

StdP: 14.39

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.9	-32.5	-43.3	0.5	-38.4	-37.1	0.7	-32.3	14.5	15.2	13.3	18.0	0.3	310	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	84.3	65.8	81.3	64.8	78.1	63.4	69.1	78.6	67.2	76.7	65.5	74.7	4.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	98.1	72.7	63.9	90.9	71.3	62.0	84.9	69.8	33.7	78.6	32.2	76.2	30.7	75.0	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.6	11.7	10.1	-41.4	90.0	7.8	2.2	-47.0	91.6	-51.6	92.9	-56.0	94.2	-61.6	95.8		
(5)			-40.3	72.0	7.4	2.3	-45.6	73.6	-50.0	74.9	-54.1	76.2	-59.6	77.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.8	-2.9	4.4	19.5	35.7	48.8	62.0	65.9	60.4	47.4	33.5	14.7	2.5
	DBStd	26.42	17.72	15.09	12.41	9.26	9.31	7.47	5.53	6.64	7.88	9.79	15.48	17.07
	HDD50	7633	1640	1276	945	435	140	6	0	6	141	512	1060	1472
	HDD65	11916	2105	1696	1410	879	508	140	57	171	528	975	1510	1937
	CDD50	1362	0	0	0	6	102	368	492	328	64	2	0	0
	CDD65	169	0	0	0	0	5	52	83	28	1	0	0	0
	CDH74	1707	0	0	0	3	84	622	688	287	23	0	0	0
	CDH80	395	0	0	0	0	14	164	152	62	3	0	0	0
Wind	WSAvg	4.0	3.2	3.6	4.5	5.2	5.3	3.8	3.3	3.3	3.8	4.4	4.2	3.6
Precipitation	PrecAvg	18.3	0.8	0.7	0.7	0.9	1.5	2.4	2.8	2.6	1.9	1.4	1.5	1.1
	PrecMax	22.8	1.6	1.5	1.7	1.7	3.7	7.5	6.1	4.7	3.5	2.2	2.4	2.1
	PrecMin	12.8	0.1	0.2	0.1	0.2	0.4	0.3	0.1	1.0	0.5	0.6	0.7	0.2
	PrecStd	2.7	0.3	0.3	0.4	0.4	0.8	1.4	1.5	1.0	0.8	0.4	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.5	36.1	48.5	67.7	82.2	88.8	88.0	85.8	77.5	61.5	45.9	35.4
		MCWB	29.2	31.3	38.9	50.3	60.3	65.1	68.2	67.2	61.0	49.3	41.1	31.5
	2%	DB	29.1	32.3	43.3	61.0	76.2	84.3	84.0	81.1	69.7	54.6	41.0	31.4
		MCWB	27.0	28.7	35.7	47.5	57.4	64.5	67.3	66.1	58.2	46.6	37.0	29.2
	5%	DB	25.8	29.5	39.8	55.8	71.2	80.9	81.0	76.9	64.9	50.5	37.4	28.3
		MCWB	23.8	26.8	33.8	44.4	55.1	63.3	66.3	63.9	55.4	45.1	34.1	26.3
	10%	DB	21.9	25.9	36.4	51.0	66.3	77.1	77.9	73.2	60.3	46.8	34.0	24.6
		MCWB	20.1	23.4	31.5	41.6	52.6	61.8	64.7	62.2	53.3	42.4	31.8	22.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.8	31.9	39.6	51.8	62.3	69.3	72.1	70.6	62.4	51.3	42.2	32.1
		MCDB	32.2	34.7	46.5	62.8	77.9	79.9	81.7	79.7	72.9	57.3	44.9	34.3
	2%	WB	27.1	29.3	36.7	48.4	59.2	67.1	69.9	67.7	59.1	48.1	37.4	29.3
		MCDB	29.1	31.9	41.7	59.2	71.6	78.7	79.4	76.8	67.8	52.8	40.5	31.3
	5%	WB	23.9	26.6	34.2	45.1	56.4	65.3	68.2	65.5	56.8	45.6	34.3	26.3
		MCDB	25.7	29.5	39.0	54.5	68.7	76.9	77.2	74.2	63.0	50.1	37.0	28.3
	10%	WB	20.2	23.3	31.8	42.0	53.7	63.6	66.4	63.4	54.2	42.5	31.8	22.8
		MCDB	22.0	25.8	36.1	50.2	65.0	74.5	74.9	71.4	59.4	46.6	33.8	24.5
Mean Daily Temperature Range	5% DB	MDBR	14.0	16.7	19.0	17.9	19.9	19.8	18.1	18.0	16.1	12.7	12.3	13.6
		MCDBR	11.5	15.1	19.8	27.1	28.4	27.4	23.4	24.4	25.3	19.5	11.8	14.9
		MCWBR	10.5	12.8	14.1	15.3	13.8	10.3	9.5	10.6	13.7	12.3	9.5	13.7
	5% WB	MCDBR	11.2	14.2	17.7	25.1	24.7	22.0	18.6	20.7	22.1	18.6	11.2	14.2
		MCWBR	10.3	12.6	13.1	15.0	13.1	9.6	8.8	10.3	12.9	12.3	9.5	13.2
Clear-Sky Solar Irradiance	taub	0.285	0.318	0.360	0.390	0.410	0.415	0.436	0.411	0.367	0.348	0.315	0.281	
	taud	1.998	1.985	1.951	2.117	2.206	2.279	2.230	2.273	2.354	2.238	2.044	1.949	
	Ebn at Noon	173	220	243	259	262	262	252	251	246	212	159	134	
	Edh at Noon	20	31	43	42	41	39	40	36	29	24	19	16	
All-Sky Solar Radiation	RadAvg	227	535	967	1387	1571	1896	1718	1352	837	425	207	130	
	RadStd	28	37	68	91	160	165	115	113	113	34	33	21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page